

week3 - Two pointers and sliding window

A. Sereja and Dima

1 second, 256 megabytes

Sereja and Dima play a game. The rules of the game are very simple. The players have n cards in a row. Each card contains a number, all numbers on the cards are distinct. The players take turns, Sereja moves first. During his turn a player can take one card: either the leftmost card in a row, or the rightmost one. The game ends when there is no more cards. The player who has the maximum sum of numbers on his cards by the end of the game, wins.

Sereja and Dima are being greedy. Each of them chooses the card with the larger number during his move.

Inna is a friend of Sereja and Dima. She knows which strategy the guys are using, so she wants to determine the final score, given the initial state of the game. Help her.

Input

The first line contains integer n ($1 \leq n \leq 1000$) — the number of cards on the table. The second line contains space-separated numbers on the cards from left to right. The numbers on the cards are distinct integers from 1 to 1000.

Output

On a single line, print two integers. The first number is the number of Sereja's points at the end of the game, the second number is the number of Dima's points at the end of the game.

input
4 4 1 2 10
output
12 5

input
7 1 2 3 4 5 6 7
output
16 12

In the first sample Sereja will take cards with numbers 10 and 2, so Sereja's sum is 12. Dima will take cards with numbers 4 and 1, so Dima's sum is 5.

B. Books

2 seconds, 256 megabytes

When Valera has got some free time, he goes to the library to read some books. Today he's got t free minutes to read. That's why Valera took n books in the library and for each book he estimated the time he is going to need to read it. Let's number the books by integers from 1 to n . Valera needs a_i minutes to read the i -th book.

Valera decided to choose an arbitrary book with number i and read the books one by one, starting from this book. In other words, he will first read book number i , then book number $i + 1$, then book number $i + 2$ and so on. He continues the process until he either runs out of the free time or finishes reading the n -th book. Valera reads each book up to the end, that is, he doesn't start reading the book if he doesn't have enough free time to finish reading it.

Print the maximum number of books Valera can read.

Input

The first line contains two integers n and t ($1 \leq n \leq 10^5$; $1 \leq t \leq 10^9$) — the number of books and the number of free minutes Valera's got. The second line contains a sequence of n integers $a_1, a_2, ..., a_n$ ($1 \leq a_i \leq 10^4$), where number a_i shows the number of minutes that the boy needs to read the i -th book.

Output

Print a single integer — the maximum number of books Valera can read.

input
4 5 3 1 2 1
output
3

input
3 3 2 2 3
output
1

C. Vasya and String

1 second, 256 megabytes

High school student Vasya got a string of length n as a birthday present. This string consists of letters 'a' and 'b' only. Vasya denotes *beauty* of the string as the maximum length of a **substring** (consecutive subsequence) consisting of equal letters.

Vasya can change no more than k characters of the original string. What is the maximum beauty of the string he can achieve?

Input

The first line of the input contains two integers n and k ($1 \leq n \leq 100\,000$, $0 \leq k \leq n$) — the length of the string and the maximum number of characters to change.

The second line contains the string, consisting of letters 'a' and 'b' only.

Output

Print the only integer — the maximum beauty of the string Vasya can achieve by changing no more than k characters.

input
4 2 abba
output
4

input
8 1 aabaabaa
output
5

In the first sample, Vasya can obtain both strings "aaaa" and "bbbb".

In the second sample, the optimal answer is obtained with the string "aaaaabaa" or with the string "aabaaaaa".

D. They Are Everywhere

2 seconds, 256 megabytes

Sergei B., the young coach of Pokemons, has found the big house which consists of n flats ordered in a row from left to right. It is possible to enter each flat from the street. It is possible to go out from each flat. Also, each flat is connected with the flat to the left and the flat to the right. Flat number 1 is only connected with the flat number 2 and the flat number n is only connected with the flat number $n - 1$.

There is exactly one Pokemon of some type in each of these flats. Sergei B. asked residents of the house to let him enter their flats in order to catch Pokemons. After consulting the residents of the house decided to let Sergei B. enter one flat from the street, visit several flats and then go out from some flat. But they won't let him visit the same flat more than once.

Sergei B. was very pleased, and now he wants to visit as few flats as possible in order to collect Pokemons of all types that appear in this house. Your task is to help him and determine this minimum number of flats he has to visit.

Input

The first line contains the integer n ($1 \leq n \leq 100\,000$) — the number of flats in the house.

The second line contains the row s with the length n , it consists of uppercase and lowercase letters of English alphabet, the i -th letter equals the type of Pokemon, which is in the flat number i .

Output

Print the minimum number of flats which Sergei B. should visit in order to catch Pokemons of all types which there are in the house.

input
3 AaA
output
2

input
7 bcAAcbcb
output
3

input
6 aaBCCe
output
5

In the first test Sergei B. can begin, for example, from the flat number 1 and end in the flat number 2.

In the second test Sergei B. can begin, for example, from the flat number 4 and end in the flat number 6.

In the third test Sergei B. must begin from the flat number 2 and end in the flat number 6.

E. Cellular Network

3 seconds, 256 megabytes

You are given n points on the straight line — the positions (x -coordinates) of the cities and m points on the same line — the positions (x -coordinates) of the cellular towers. All towers work in the same way — they provide cellular network for all cities, which are located at the distance which is no more than r from this tower.

Your task is to find minimal r that each city has been provided by cellular network, i.e. for each city there is at least one cellular tower at the distance which is no more than r .

If $r = 0$ then a tower provides cellular network only for the point where it is located. One tower can provide cellular network for any number of cities, but all these cities must be at the distance which is no more than r from this tower.

Input

The first line contains two positive integers n and m ($1 \leq n, m \leq 10^5$) — the number of cities and the number of cellular towers.

The second line contains a sequence of n integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$) — the coordinates of cities. It is allowed that there are any number of cities in the same point. All coordinates a_i are given in non-decreasing order.

The third line contains a sequence of m integers $b_1, b_2, \dots, b_m$ ($-10^9 \leq b_j \leq 10^9$) — the coordinates of cellular towers. It is allowed that there are any number of towers in the same point. All coordinates b_j are given in non-decreasing order.

Output

Print minimal r so that each city will be covered by cellular network.

input
3 2 -2 2 4 -3 0
output
4

input
5 3 1 5 10 14 17 4 11 15
output
3

F. Petya and Strings

2 seconds, 256 megabytes

Little Petya loves presents. His mum bought him two strings of the same size for his birthday. The strings consist of uppercase and lowercase Latin letters. Now Petya wants to compare those two strings lexicographically. The letters' case does not matter, that is an uppercase letter is considered equivalent to the corresponding lowercase letter. Help Petya perform the comparison.

Input

Each of the first two lines contains a bought string. The strings' lengths range from 1 to 100 inclusive. It is guaranteed that the strings are of the same length and also consist of uppercase and lowercase Latin letters.

Output

If the first string is less than the second one, print "-1". If the second string is less than the first one, print "1". If the strings are equal, print "0". Note that the letters' case is not taken into consideration when the strings are compared.

input
aaaa aaaA
output
0

input
abs Abz
output
-1

input
abcdefg AbCdEfF
output
1

If you want more formal information about the lexicographical order (also known as the "dictionary order" or "alphabetical order"), you can visit the following site:

- http://en.wikipedia.org/wiki/Lexicographical_order

G. Gravity Flip

1 second, 256 megabytes

Little Chris is bored during his physics lessons (too easy), so he has built a toy box to keep himself occupied. The box is special, since it has the ability to change gravity.

There are n columns of toy cubes in the box arranged in a line. The i -th column contains a_i cubes. At first, the gravity in the box is pulling the cubes downwards. When Chris switches the gravity, it begins to pull all the cubes to the right side of the box. The figure shows the initial and final configurations of the cubes in the box: the cubes that have changed their position are highlighted with orange.



Given the initial configuration of the toy cubes in the box, find the amounts of cubes in each of the n columns after the gravity switch!

Input

The first line of input contains an integer n ($1 \leq n \leq 100$), the number of the columns in the box. The next line contains n space-separated integer numbers. The i -th number a_i ($1 \leq a_i \leq 100$) denotes the number of cubes in the i -th column.

Output

Output n integer numbers separated by spaces, where the i -th number is the amount of cubes in the i -th column after the gravity switch.

input
4 3 2 1 2
output
1 2 2 3

input
3 2 3 8
output
2 3 8

The first example case is shown on the figure. The top cube of the first column falls to the top of the last column; the top cube of the second column falls to the top of the third column; the middle cube of the first column falls to the top of the second column.

In the second example case the gravity switch does not change the heights of the columns.

H. Anton and Danik

1 second, 256 megabytes

Anton likes to play chess, and so does his friend Danik.

Once they have played n games in a row. For each game it's known who was the winner — Anton or Danik. None of the games ended with a tie.

Now Anton wonders, who won more games, he or Danik? Help him determine this.

Input

The first line of the input contains a single integer n ($1 \leq n \leq 100\,000$) — the number of games played.

The second line contains a string s , consisting of n uppercase English letters 'A' and 'D' — the outcome of each of the games. The i -th character of the string is equal to 'A' if the Anton won the i -th game and 'D' if Danik won the i -th game.

Output

If Anton won more games than Danik, print "Anton" (without quotes) in the only line of the output.

If Danik won more games than Anton, print "Danik" (without quotes) in the only line of the output.

If Anton and Danik won the same number of games, print "Friendship" (without quotes).

input
6 ADAAAA
output
Anton

input
7 DDDAADA
output
Danik

input
6 DADADA
output
Friendship

In the first sample, Anton won 6 games, while Danik — only 1. Hence, the answer is "Anton".

In the second sample, Anton won 3 games and Danik won 4 games, so the answer is "Danik".

In the third sample, both Anton and Danik won 3 games and the answer is "Friendship".

I. Vanya and Fence

1 second, 256 megabytes

Vanya and his friends are walking along the fence of height h and they do not want the guard to notice them. In order to achieve this the height of each of the friends should not exceed h . If the height of some person is greater than h he can bend down and then he surely won't be noticed by the guard. The height of the i -th person is equal to a_i .

Consider the width of the person walking as usual to be equal to 1, while the width of the bent person is equal to 2. Friends want to talk to each other while walking, so they would like to walk in a single row. What is the minimum width of the road, such that friends can walk in a row and remain unattended by the guard?

Input

The first line of the input contains two integers n and h ($1 \leq n \leq 1000$, $1 \leq h \leq 1000$) — the number of friends and the height of the fence, respectively.

The second line contains n integers a_i ($1 \leq a_i \leq 2h$), the i -th of them is equal to the height of the i -th person.

Output

Print a single integer — the minimum possible valid width of the road.

input
3 7 4 5 14
output
4

input
6 1 1 1 1 1 1 1
output
6

input
6 5 7 6 8 9 10 5
output
11

In the first sample, only person number 3 must bend down, so the required width is equal to $1 + 1 + 2 = 4$.

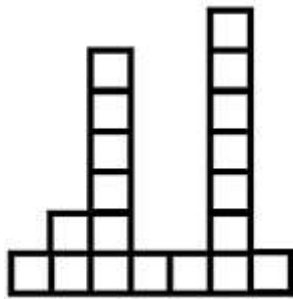
In the second sample, all friends are short enough and no one has to bend, so the width $1 + 1 + 1 + 1 + 1 + 1 = 6$ is enough.

In the third sample, all the persons have to bend, except the last one. The required minimum width of the road is equal to $2 + 2 + 2 + 2 + 2 + 1 = 11$.

J. Fence

1 second, 256 megabytes

There is a fence in front of Polycarpus's home. The fence consists of n planks of the same width which go one after another from left to right. The height of the i -th plank is h_i meters, distinct planks can have distinct heights.



Fence for $n = 7$ and $h = [1, 2, 6, 1, 1, 7, 1]$

Polycarpus has bought a posh piano and is thinking about how to get it into the house. In order to carry out his plan, he needs to take exactly k consecutive planks from the fence. Higher planks are harder to tear off the fence, so Polycarpus wants to find such k consecutive planks that the sum of their heights is minimal possible.

Write the program that finds the indexes of k consecutive planks with minimal total height. Pay attention, the fence is not around Polycarpus's home, it is in front of home (in other words, the fence isn't cyclic).

Input

The first line of the input contains integers n and k ($1 \leq n \leq 1.5 \cdot 10^5, 1 \leq k \leq n$) — the number of planks in the fence and the width of the hole for the piano. The second line contains the sequence of integers $h_1, h_2, ..., h_n$ ($1 \leq h_i \leq 100$), where h_i is the height of the i -th plank of the fence.

Output

Print such integer j that the sum of the heights of planks $j, j + 1, ..., j + k - 1$ is the minimum possible. If there are multiple such j 's, print any of them.

input
7 3 1 2 6 1 1 7 1
output
3

In the sample, your task is to find three consecutive planks with the minimum sum of heights. In the given case three planks with indexes 3, 4 and 5 have the required attribute, their total height is 8.